

CTF CHALLENGE

Cryptography · Hard Difficulty · 500 pts

Challenge Name	The Registrar's Index	Category	Cryptography
Difficulty	Hard (500 pts)	Flag Format	flag{...}
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Challenge

■ PARTICIPANT INSTRUCTIONS

A university registrar disappeared, leaving behind only an encrypted string and a partial database export — eight student records, each with an ID and a score.

Somewhere in those numbers lies a hidden key. Use it to decrypt the encoded string below and recover the flag.

Two reference tables have been provided with this challenge: a Residue Lookup Table and a Cipher Alphabet. Both are required for the solution.

No external tools or databases are needed. **All data required for the solution is on this page.**

Student Records (Database Export)

#	Student ID	Score
1	1000158	211
2	1000148	271
3	1000220	307
4	1000072	349
5	1000189	379
6	1000272	439
7	1000379	457
8	1000374	457

Encoded String

```
EEHI{DF4R1HJ_0M_WX3_P1MB3Y}
```

Note: This string uses a custom encoding scheme. Standard decoders will not work.

Residue Lookup Table (Index → Character)

Each index (0–96) maps to exactly one character. Use this table to convert computed residue values into characters.

+0	+1	+2	+3	+4	+5	+6	+7	+8	+9
0:>	1:8	2::	3:t	4:i	5:X	6:<	7:c	8:S	9:l
10:9	11:Z	12:p	13:w	14:h	15:4	16:&	17:N	18:G	19:6
20:2	21:l	22:*	23:]	24:C	25:{	26:'	27:(	28:u	29:[
30:B	31:#	32:K	33:Y	34:O	35:o	36:n	37:5	38:U	39:W
40:I	41:M	42:A	43:y	44:`	45:R	46:f	47:a	48:~	49:L
50:V	51:v	52:J	53:g	54:,	55:F	56:Q	57:)	58:?	59:^
60:%	61:b	62:!	63:T	64:z	65:_	66:/	67:	68:H	69:3
70:\$	71:s	72:x	73:E	74:k	75:r	76:+	77:D	78:@	79:7
80:=	81:j	82:;	83:d	84:q	85:"	86:e	87:m	88:}	89:P
90:\	91:-	92:0	93:.	94:	95:.	96:!			

Cipher Alphabet

This alphabet replaces the standard A–Z ordering for all encryption operations in this challenge.

Standard (A–Z)	Cipher Alphabet
ABCDEFGHIJKLMNOPQRSTUVWXYZ	ZECNRKIJPTBUAFGSXMVYDWOLHQ

Hints

Hint 1 (–50 pts)

The student records are not decoration. Each row contains two numbers that together produce a single character when processed correctly.

Hint 2 (-100 pts)

The row number itself is part of the calculation. The result of your calculation is an index into the Residue Lookup Table.

Hint 3 (-150 pts)

For row i : compute $(\text{Student_ID} \times i) \bmod \text{Score}$. Look up the result in the Residue Lookup Table to get a character. The 8 characters form a key for a substitution cipher that uses the Cipher Alphabet above.